SME Finance Copilot: Engineering & Business Guide

From Zero-Jargon First Principles to Autonomous Financial Agent Architecture

2026 FinTechathon Track E | Host: Tsinghua University PBCSF & WeBank (微众银行) | Deadline: Oct 20, 2026

1. Executive Overview & The FinTechathon Challenge

The theme of the 2026 8th FinTechathon International Track is **'AI as a Financial Participant'**. Unlike conversational chatbots that passively summarize documents, an autonomous financial participant actively monitors real-time financial telemetry, projects cash runway, formulates actionable financing solutions, and executes within banking sandboxes under strict permission guardrails.

The Official Prompt (Topic E): 'A finance copilot for small-business owners, offering financing-option matching, cash-flow management with alerts, and business financial analysis.'

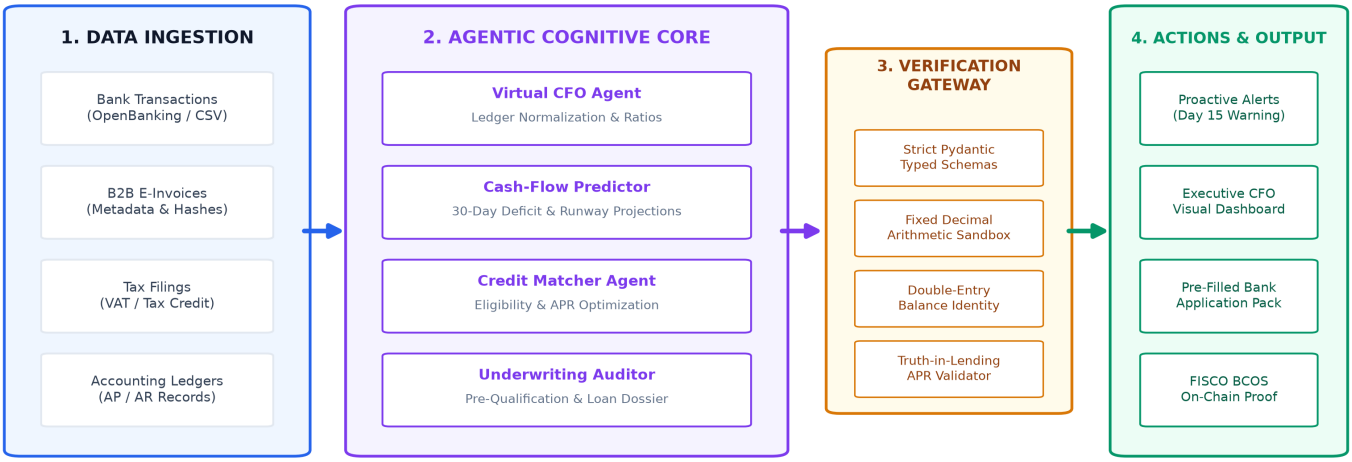


Figure 1: End-to-End Multi-Agent Architecture for SME Finance Copilot

Evaluation Pillar	Weight	Evaluation Focus & Scoring Strategy
Task Completion	40%	End-to-end execution of all 3 sub-capabilities: transaction ingestion, deterministic cash-flow projections with proactive alert triggers, and loan option matching with sandbox bank interaction.
Security & Compliance	30%	Three-tier permission model, formal separation of LLM reasoning from mathematical calculations, PII redaction, human-in-the-loop (HITL) authorization, and security self-assessment report.
Innovation & Interaction	30%	Proactive agent behavior (initiating alerts without human prompts), multi-scenario forecasting, and interactive generative UI dashboards.
On-Chain Challenge Bonus	0–5 Pts	Consortium blockchain integration (FISCO BCOS) to eliminate invoice double-financing and establish an immutable audit ledger.

2. Zero-Jargon Foundation: How Small Businesses Actually Work

Finance and marketing feel overwhelmingly complex because traditional banking uses archaic jargon to describe simple concepts. As software engineers, you do not need an MBA to excel in this competition. The entire domain operates on straightforward logic.

2.1 The Phone Battery Metaphor

Think of a business's finances exactly like your smartphone battery:
• **Battery Percentage:** You check your phone and see 18% remaining. That is your current **Bank Account Balance**.
• **Drain Rate Prediction:** Your phone knows screen brightness and apps in use will drain the battery to 0% in 20 minutes. That is **Cash Runway Forecasting**.
• **Proactive Warning:** Your phone does not wait for you to ask; it triggers a popup: 'Battery below 20%. Turn on Low Power Mode?' That is **Proactive Financial Alerts**.
• **Actionable Solution:** You tap 'Yes' and the system extends battery life. That is **Financing Matching**.
Topic E is simply building the intelligent low-battery warning system for small businesses.

2.2 Meet Marco: The Shoemaker's Dilemma

Meet Marco. Marco makes high-end handmade leather shoes. He is a master craftsman who spends 12 hours a day cutting leather. He is not an accountant. When Marco opens his bank app and sees **$5,000**, he feels wealthy and safe. **Here is the trap:** Marco forgot that on Friday, employee wages of **$3,000** are due, and on Tuesday, workshop rent of **$2,500** will be auto-debited. Total upcoming obligations: **$5,500**. Next Tuesday, his account hits -$500, checks bounce, and penalty fees hit. Marco has dozens of eager customers, but his business freezes because he lost track of the timing of his bills.

2.3 The Jargon Translator: Banker Terms in Plain English

Banker Term	Plain-English Meaning	Real-World Context / Example
SME	Small & Medium Enterprise	Regular 3-to-25 person businesses: print shops, restaurants, Shopify brands.
Cash Flow	Money In vs. Money Out	Real physical dollars arriving in the bank account vs. dollars leaving for bills.
Accounts Receivable (AR)	Customer IOUs	Delivered goods for which the client has not yet paid.
Net 30 / Net 60	Delayed Payment Terms	Corporate clients legally taking 30 to 60 days to settle invoices after delivery.
Cash-Flow Crunch	Temporary Cash Shortage	Having $100 in the bank when a $1,000 payroll check is due tomorrow.
Invoice Factoring	Selling an IOU for Cash Today	Getting an instant 85% advance on an unpaid invoice from a bank for a small fee.
Underwriting	Lender Risk Check	The bank checking transaction history to ensure the borrower can repay the loan.
Financial Participant	Proactive Autonomous Agent	An AI that alerts the user first when risks emerge, rather than waiting for questions.

Core Software Logic (How an Engineer Solves This in Code)

```
current_balance = 5000
upcoming_bills = 3000 + 2500 # Payroll ($3k) + Rent ($2.5k) due in 5 days
if current_balance < upcoming_bills:
    shortfall = upcoming_bills - current_balance # Shortfall = $500
    send_proactive_alert(f'Warning: Cash shortfall of ${shortfall} detected on Day 5!')
    best_loan = match_cheapest_credit_option(shortfall, due_days=5)
    present_options_with_2fa_confirmation(best_loan)
```

3. The Working Capital Trap: Why 'Profit' is an Illusion

The most critical business concept engineers must grasp is the fundamental divergence between **accrual accounting (权责发生制)** and **cash accounting (收付实现制)**. Under GAAP/IFRS standards, revenue is recognized the instant an order is delivered or an invoice is issued, even if zero dollars have been received. Expenses are recognized when incurred. A business can report $40,000 in net profit while having $0 in its bank account.

3.1 The Marketing & Growth Trap

Beginners often think that when a business struggles, it simply needs more marketing. In reality, **scaling sales without financing is lethal:**

1. An apparel company spends $10,000 on TikTok ads. The ad platform charges the credit card **immediately**.
2. The campaign goes viral! 500 bulk orders arrive, worth $100,000 in revenue.
3. To fulfill those orders, the company must purchase $50,000 of blank garments and pay $20,000 in labor **today**.
4. Corporate buyers stipulate **Net 60 terms**; payment will only arrive in two months.
5. If the business lacks $70,000 in liquid capital today, it defaults on production, breaks contracts, and goes bankrupt. **Growth consumes cash.**

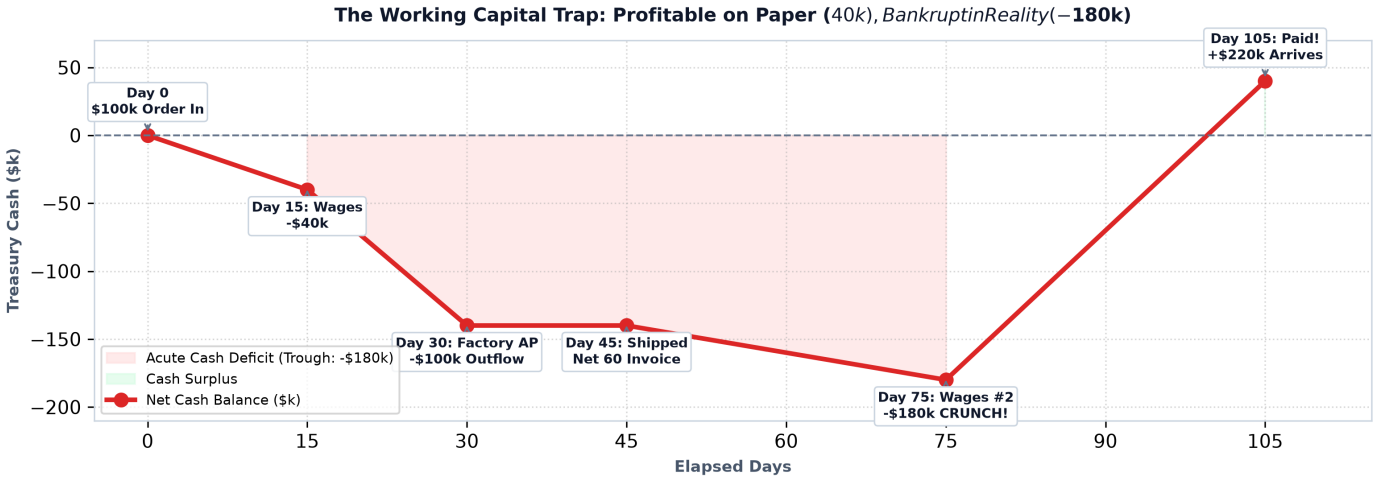


Figure 2: The Anatomy of a Working Capital Crunch: Accrual Profit vs. Cash Deficit Timeline

3.2 The Cash Conversion Cycle (CCC / 现金循环周期)

The timeline above illustrates the mathematical formula behind cash deficits: **CCC = DIO + DSO - DPO**.

• **DIO (Days Inventory Outstanding):** (Average Inventory / COGS) * 365. How long cash is trapped as physical inventory.
• **DSO (Days Sales Outstanding):** (Accounts Receivable / Total Credit Sales) * 365. Average days clients take to pay invoices.
• **DPO (Days Payables Outstanding):** (Accounts Payable / COGS) * 365. Average days the business delays paying its own vendors.

The Deficit Trough: In the scenario above, despite an eventual $40,000 accounting profit, the company endures an acute **-$180,000 cash deficit** between Days 30 and 104. The Copilot's mission is to identify this trough before it happens and bridge it safely.

4. Deconstructing the 3 Pillars & Deterministic Financial Core

Topic E specifies three explicit core capabilities. Here is how they operate inside the autonomous copilot:

- **Pillar 1: Business Financial Analysis (The Diagnostic Doctor):** Ingests raw, unstructured bank CSVs and invoices. Automatically cleans and categorizes transaction lines into standard chart-of-accounts. Evaluates gross margins, expense leaks, and working capital ratios.
- **Pillar 2: Cash-Flow Management with Alerts (The Smoke Detector):** Projects future bank balances across a 30-day sliding window. Detects when recurring expenses (rent, payroll, tax auto-debits) will cause an overdraft, triggering proactive warnings before checks bounce.
- **Pillar 3: Financing-Option Matching (The Unbiased Broker):** Scans available digital financial products (invoice factoring, revolving credit, PO financing). Ranks options by total effective cost and pre-fills application dossiers for instant execution.

4.1 Deterministic Financial Core: Strict Anti-Hallucination Math

The Copilot must **never calculate financial metrics using LLM token generation**. All arithmetic must be executed in isolated Python kernels using decimal.Decimal to guarantee exact precision:

Financial Metric	Deterministic Formula	Operational Benchmark & Underwriter Threshold
DSCR (Debt Service Coverage)	Net Operating Income / Total Debt Service	**≥ 1.25x:** Healthy capacity. **1.0–1.20x:** Vulnerable. **< 1.0x:** Immediate default risk.
Quick Ratio (Acid-Test Ratio)	(Cash + Equivalents + AR) / Current Liabilities	**≥ 1.0x:** High liquidity buffer. **< 0.8x:** High risk if any customer delays payment.
Current Ratio	Total Current Assets / Current Liabilities	**1.5x–2.0x:** Standard commercial healthy range. **< 1.0x:** Negative working capital.
OCF Margin (Operating Cash Flow)	(Operating Cash Flow / Revenue) * 100%	**> 15%:** Excellent cash conversion. **≤ 0%:** Business burns real cash on every unit sold.
Cash Runway	Available Cash Balance / Monthly Net Burn	**> 6 Mos:** Secure. **2–3 Mos:** Urgent financing alert. **< 30 Days:** Immediate liquidity crisis.

The Hackathon Demo Scenario: The Story of Alex's Apparel Co.

• **Day 1:** Alex receives a $50,000 order for 2,000 jackets from a major conference. Payment is Net 45.
• **Day 5:** Alex spends $25,000 on raw fabrics. Bank balance drops from $32,000 to $7,000.
• **Day 15:** Payroll of $12,000 is due in 48 hours. Alex only has $7,000 in cash ($5,000 shortfall).
• **The Copilot Action:** Detected on Day 5 $\to$ matched an 8% APR purchase order facility $\to$ pre-filled loan dossier $\to$ 2FA confirmed $\to$ funds arrived on Day 12. Payroll cleared without incident.

5. Digital SME Underwriting & Financing Instruments

Commercial banks historically rejected micro and small businesses due to the lack of audited financial statements, credit ratings, and physical real estate collateral. Modern digital lenders (such as **WeBank** in China and **Stripe Capital** or **Brex** globally) replace audited books with programmatic ingestion of real-time operational and cash-flow telemetry:

- **Daily Average Balance (DAB) & Volatility:** Lenders evaluate sliding-window 30/60/90-day balances. Coefficient of variation (CV = std_dev / mean) measures cash stability. Erratic balances reduce credit limits.
- **NSF (Non-Sufficient Funds) & Overdraft Velocity:** Bounced checks, failed ACH transfers, or negative intraday balances are acute red flags. More than 1–2 NSF occurrences within 90 days triggers automated underwriting rejection.
- **Customer Concentration (HHI Index):** Herfindahl-Hirschman Index HHI = sum(s_k^2) on incoming client revenue. If HHI > 0.25, the business is overly dependent on 1–2 enterprise buyers; a single dispute can freeze total debt service.
- **Tax Data Linkage ('银税互动' / Bank-Tax Interaction):** Formal corporate tax filing records serve as an unalterable proxy for commercial vitality. Underwriters analyze tax compliance ratings (Tier A/B/M), quarterly VAT trends, and consistency between tax declarations and bank account deposits.
- **B2B VAT E-Invoices:** Electronic invoice streams verify real business background. Key anomaly metrics include invoice cancellation rates and red-letter invoice ratios (indicating commercial disputes or synthetic transaction fabrication).

5.1 Comparison of SME Financing Instruments

Financing Instrument	Collateral & Structure	Pricing Metric (APR vs Factor)	Ideal Scenario for SME
Revolving Credit Line (Working Capital Facility)	Unsecured or corporate guarantee. Borrow and repay on demand.	**5% – 12% APR** Simple interest on drawn capital.	Seasonal liquidity buffers, short-term inventory procurement.
Invoice Factoring (Accounts Receivable Discounting)	Pledged B2B invoices. 80–90% upfront cash advance.	**1% – 3% per 30 days** (Effective APR: 15% – 36%+).	Long payment terms (Net 60/90) with high-credit corporate buyers.
Purchase Order Financing (Pre-Shipment Supplier Advance)	Confirmed customer PO. Lender pays supplier directly.	**2% – 5% flat fee** (Effective APR: 24% – 50%+).	Pre-manufacturing capital when receiving large enterprise purchase orders.
Revenue-Based Financing (RBF / MCA Advance)	Repayment via fixed % of daily/monthly card sales.	**Factor Rate: 1.15x – 1.35x** (Effective APR: 35% – 80%+).	E-commerce / POS retail with high margins and volatile seasonal revenue.

Truth-In-Lending Guardrail: Factor Rate to APR Conversion

Factor rates multiply the total initial advance rather than declining principal. The Copilot must always calculate and display the standardized annualized APR for transparent disclosure:
APR ≈ ((Factor Rate - 1) / Repayment Term Days) * 365 * 100%
Example: A $50,000 cash advance with a 1.20 factor rate ($60,000 total payback) repaid over 90 days incurs an effective **APR of 81.1%**. The Copilot protects owners by alerting them when an online loan offer carries predatory effective rates.

6. WeBank Strategy & The On-Chain Bonus (+5 Points)

To win high praise from WeBank executive judges, your team must align with their corporate heritage and technical foundations.

6.1 WeBank's 'Weiyidai' (微业贷) Paradigm

In China, SMEs represent the **'56789' phenomenon** (50% of tax, 60% of GDP, 70% of tech innovation, 80% of urban jobs, 90% of business entities). Historically, commercial banks allocated less than 20% of loans to SMEs. In 2017, WeBank revolutionized the sector with **Weiyidai (微业贷)**:

- **The '321' Experience:** 3 minutes to submit mobile application, 2 seconds for algorithmic underwriting decision, 1-click instant drawdown, 0 human loan officers involved.
- **The 'A+B' Dual-Engine Risk Model:** Seamlessly fuses **Dimension A (Corporate Telemetry)**—including provincial tax databases ('银税互动'), corporate legal filings, and utility bills—with **Dimension B (Owner Persona)**—including personal credit bureau scoring, biometric anti-spoofing, and counterparty graph analysis.

6.2 Capturing the On-Chain Bonus with FISCO BCOS

The competition offers a **0–5 point Challenge Bonus** for on-chain exploration. In Topic E, consortium blockchain is not a gimmick—it solves the single greatest vulnerability in SME trade finance: **Invoice Double-Financing (发票重复融资 / 重复质押)**.

The Double-Financing Threat: An SME issues an electronic invoice for $300,000 to an enterprise buyer. It then presents identical copies of the invoice to Bank A, Bank B, and Factor C, receiving $900,000 in advances against $300,000 of real assets.

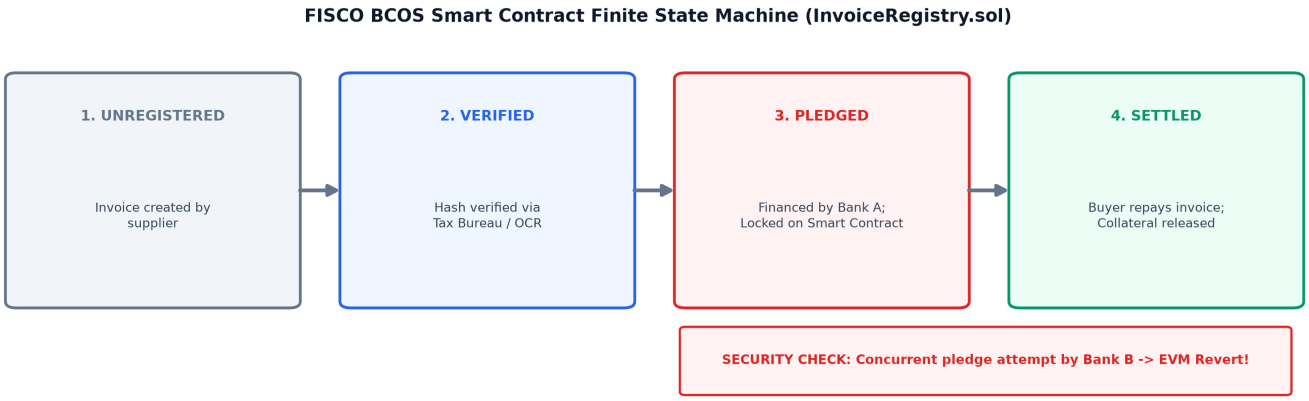


Figure 3: FISCO BCOS Smart Contract Finite State Machine (InvoiceRegistry.sol)

FISCO BCOS Smart Contract Architecture: InvoiceRegistry.sol

1. Deterministic Invoice Fingerprint:
bytes32 invoiceHash = keccak256(abi.encodePacked(taxId, invoiceNumber, amount, issueDate));
2. Atomic Finite State Machine:
enum InvoiceState { UNREGISTERED, VERIFIED, PLEDGED, SETTLED }
3. Multi-Institutional Locking: When a partner bank initiates factoring, the smart contract transitions the invoice to PLEDGED and records the lending institution's address. Any concurrent financing attempt by another bank is automatically reverted by consensus logic.
4. Tamper-Proof Audit Anchor: The AI Copilot periodically anchors a Merkle root of all internal cash-flow projections and financial recommendations to the blockchain, creating an immutable compliance ledger.

7. Three-Tier Permission Security Model & Compliance (30% Score)

FinTechathon explicitly grades **Security & Compliance at 30%** and requires a mandatory **Security Self-Assessment Report**. Your system must enforce a formal permission-tier architecture separating read-only telemetry, proposal drafting, and execution.



Figure 4: Three-Tier Defense-in-Depth Security Matrix for Autonomous Financial Agents

Permission Tier	Operational Scope	Permitted Autonomous Actions	Security & Verification Boundary
Tier 1: Observation	Telemetry Ingestion & Financial Analysis	Syncing bank transactions, parsing invoice PDFs, computing DSCR/Runway, running Monte Carlo cash-flow forecasts.	**Zero Mutation:** Read-only scoped API tokens. Air-gapped from transactional endpoints. Zero PII stored in raw LLM prompts.
Tier 2: Staging & Drafting	Proposal Formulation & Pre-Underwriting	Drafting credit line drawdowns, matching factoring products, staging supplier payment batches, compiling loan dossiers.	**Immutable Intent Store:** Generates signed ActionIntent JSON objects with idempotency keys. Staged in queue; strictly non-executable.
Tier 3: High-Stakes Action	Capital Movement & Binding Contracts	Executing wire transfers, signing bank loan agreements, pledging on-chain invoice tokens, drawing down credit lines.	**Strict HITL + 2FA:** Blocked until verified by owner via out-of-band FIDO2/WebAuthn biometrics or TOTP. Multi-sig execution quorum.

8. Prototype Demo Recipe & Official Submission Checklist

To win top marks in the 10-minute presentation and review, design your prototype around 3 interactive tabs:

- **Tab 1: Virtual CFO Dashboard:** Real-time metric cards (Runway: 18 Days, Quick Ratio: 0.75 Warning, Burn Rate: $12k/mo) and an interactive cash-flow timeline slider showing the Day 15 deficit.
- **Tab 2: Proactive Agent Feed:** Displays autonomous agent notifications (e.g. 'Liquidity Warning: Payroll on Day 15 will cause a $5,000 deficit. Here are 3 matched credit options').
- **Tab 3: Financing Matcher & 2FA Execution:** Transparent APR side-by-side comparison, auto-compiled loan dossier, and a 2FA confirmation pop-up proving Human-in-the-Loop governance.

8.1 Official Submission Checklist (Deadline: Oct 20, 2026, 23:59 Beijing Time)

Deliverable	Format & Constraints	Key Content to Win Top Marks
01. Technical Doc	PDF / Word Document	System architecture diagram, agent state machine (LangGraph), deterministic formulas, API specs, and on-chain contract design.
02. Presentation Deck	PDF / PPT (Max 10 mins)	Problem statement ('The Story of Alex'), solution demo screenshots, technical novelty, WeBank alignment, and market viability.
03. Demo Video	Video (Optional, ≤ 5 mins)	End-to-end screen recording: Ingesting bank CSV → Detecting Day 18 cash crunch → Alerting owner → Matching factoring → HITL 2FA drawdown.
04. Source Code	GitHub Repo + README	Clean repository with Docker Compose / one-line start, deployment documentation, unit test suites (>80% coverage), and mock seed data.
05. Security Report	PDF / Markdown Report	Three-tier permission table, known risk register (prompt injection, PII leak, arithmetic error), and formal defense implementation details.
06. Execution Evidence	Logs & On-Chain Tx Hashes	Recorded sandbox operation logs demonstrating loan pre-qualification API calls and verifiable on-chain transaction hashes for invoice registration.

8.2 Suggested Technology Stack

- **Backend & Multi-Agent Framework:** Python 3.11+, **FastAPI**, and **LangGraph** (for stateful multi-agent cycles: Ingestion Agent → Analyst Agent → Matcher Agent → Security Gatekeeper).
- **Deterministic Math Engine:** Python decimal.Decimal, Pydantic v2 schemas with strict invariant validation.
- **Frontend & Generative UI: Next.js 14+** (App Router, Tailwind CSS, shadcn/ui, Recharts) with dynamic interactive cash-flow timeline sliders and scenario simulators.
- **Consortium Blockchain (Bonus): FISCO BCOS** (or EVM testnet with Solidity 0.8.20) for InvoiceRegistry.sol smart contract and verifiable event logs.

Summary Recommendation for the Team: Focus on demonstrating the complete, proactive crisis story of Alex's Apparel Co. Showcase how the AI Copilot monitors telemetry in the background, warns the owner 10 days before an overdraft occurs, protects them from predatory high-interest debt, secures the loan on-chain to prevent fraud, and executes safely with biometric Human-in-the-Loop verification. This narrative and architecture addresses 100% of the competition's scoring criteria and positions your team as a front-runner for the top prize.